Activity 5 — ATM Transaction Sequence Simulation

1. Problem Statement

Develop a Python program to simulate a simple ATM transaction. The program allows a user to check their balance, deposit money, and withdraw money while maintaining the correct account balance.

2. Objectives

- To simulate basic ATM operations.
- To calculate the updated account balance.
- To check whether sufficient balance is available for withdrawal.
- To understand Python conditional statements and arithmetic operations.

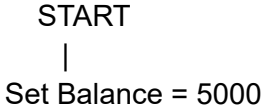
3. Assumptions

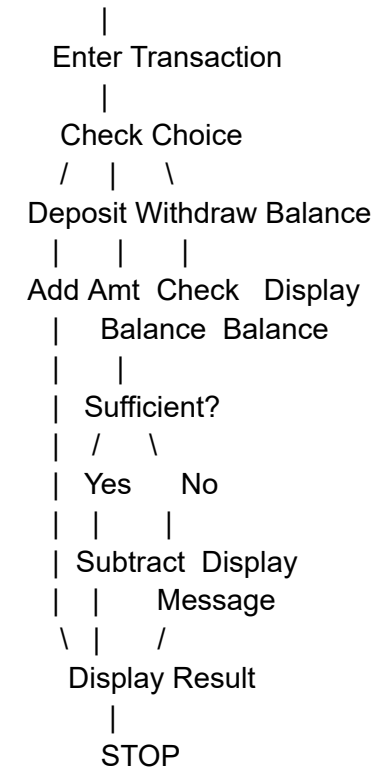
- Initial account balance = ₹5,000.
- The user can deposit or withdraw money.
- The transaction amount must be positive.
- A withdrawal cannot exceed the available balance.
- This is a basic educational simulation, not a real banking system.

4. Algorithm

1. Start.
2. Set the initial balance to ₹5,000.
3. Display the options: Check Balance, Deposit, and Withdraw.
4. Read the user's choice.
5. If the choice is Deposit, add the amount to the balance.
6. If the choice is Withdraw, check the balance and subtract the amount if sufficient.
7. If the choice is Check Balance, display the current balance.
8. Display the result.
9. Stop.

5. Flowchart





6. Pseudocode

```
START
balance = 5000

INPUT choice

IF choice = 1
   INPUT amount
   IF amount > 0
      balance = balance + amount
      PRINT balance
   ELSE
      PRINT "Invalid amount"
   ENDIF

ELSE IF choice = 2
   INPUT amount
   IF amount <= 0
      PRINT "Invalid amount"
   ELSE IF amount <= balance
      balance = balance - amount
      PRINT balance
   ELSE
      PRINT "Insufficient balance"
   ENDIF
```

```
ELSE IF choice = 3
   PRINT balance

ELSE
   PRINT "Invalid choice"
ENDIF

STOP
```

7. Python Source Code

File name: `activity13.py`

```
balance = 5000

print("1. Deposit")
print("2. Withdraw")
print("3. Check Balance")

choice = int(input("Enter your choice: "))

if choice == 1:
    amount = float(input("Enter deposit amount: "))

    if amount > 0:
        balance = balance + amount
        print("Updated Balance: Rs.", balance)
    else:
        print("Invalid amount")

elif choice == 2:
    amount = float(input("Enter withdrawal amount: "))

    if amount <= 0:
        print("Invalid amount")
    elif amount <= balance:
        balance = balance - amount
        print("Updated Balance: Rs.", balance)
    else:
        print("Insufficient balance")

elif choice == 3:
    print("Current Balance: Rs.", balance)

else:
    print("Invalid choice")
```

8. Test Cases

Choice	Amount	Expected Output
Deposit	₹1,000	Updated balance: ₹6,000
Withdraw	₹2,000	Updated balance: ₹3,000
Withdraw	₹6,000	Insufficient balance
Check Balance	—	Current balance: ₹5,000
Deposit	₹-500	Invalid amount

Each test starts with the initial balance of ₹5,000.

9. Output Screenshot and Conclusion

Sample Output:

1. Deposit
2. Withdraw
3. Check Balance
Enter your choice: 1
Enter deposit amount: 1000
Updated Balance: Rs. 6000.0

Output Screenshot: Run `activity5.py` in Python IDLE or VS Code. Capture the program inputs and output, and save the screenshot as `activity5_output.png`.

Conclusion: This program simulates basic ATM transactions using Python variables, arithmetic operators and conditional statements.

Input code:

```python
balance = 5000

print("1. Deposit")
print("2. Withdraw")
print("3. Check Balance")

choice = int(input("Enter your choice: "))

if choice == 1:
    amount = float(input("Enter deposit amount: "))

    if amount > 0:
        balance = balance + amount
        print("Updated Balance: Rs.", balance)
    else:
        print("Invalid amount")

elif choice == 2:
    amount = float(input("Enter withdrawal amount: "))

    if amount <= 0:
        print("Invalid amount")
    elif amount <= balance:
        balance = balance - amount
        print("Updated Balance: Rs.", balance)
    else:
        print("Insufficient balance")

elif choice == 3:
    print("Current Balance: Rs.", balance)

else:
    print("Invalid choice")
```

OUTPUT:

```
1. Deposit
2. Withdraw
3. Check Balance
Enter your choice: 1
Enter deposit amount: 1000
Updated Balance: Rs. 6000.0

==== RESTART: C:/Users/phani/Desktop/ATM T
1. Deposit
2. Withdraw
3. Check Balance
Enter your choice: 2
Enter withdrawal amount: 2000
Updated Balance: Rs. 3000.0

=================================== RESTAR
1. Deposit
2. Withdraw
3. Check Balance
Enter your choice: 2
Enter withdrawal amount: 6000
Insufficient balance

=================================== RESTAR
1. Deposit
2. Withdraw
3. Check Balance
Enter your choice: 3
Current Balance: Rs. 5000

=================================== RESTAR
1. Deposit
2. Withdraw
3. Check Balance
Enter your choice: 1
Enter deposit amount: -500
Invalid amount
```